

A tweet sentiment classification approach using an ensemble classifier

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ABSTRACT

Social media users are more receptive to products or events and share their thoughts through raw textual data, which is classified as semi-structured data. This data, which is presented using a variety of terminologies, is noisy by nature but yet contains important information and superfluous details, giving analysts a way to identify patterns and knowledge. This hidden information must be extracted from language data in order to make informed decisions and create strategic plans for entering new markets. Among the most prominent fields of study are natural language processing (NLP) and data mining techniques, especially when it comes to sentiment analysis—the process of identifying the feelings and insights concealed in the data. Twitter is one of the significant microblogging platform with millions of users. These users use Twitter to share sentiments using hash tags on different topics and to make status updates known as tweets. Twitter is therefore regarded as a significant real-time source and as one of the most active opinion indicators. The volume of information is produced by Twitter is enormous and manually scanning the entire data set is difficult process. The paper proposed an ensemble classifier to categorize emotion of the tweets on the basis of polarities such as positive and negative.

In our study, we ensemble classifiers which is a combination of Random Forest (RF), Support Vector Machine (SVM) and Decision Tree (DT). The data is collected from Twitter API and the Twitter data is analysed autonomously to define public view on particular topic. The features obtained after the process of dimensionality reduction using LDA undergoes the stage of feature selection using Wrapper based technique. The iterative Wrapper based technique predict score for the features, the features with low score are ignored and high score is proceeded for classification. The ensemble classifier used Adaptive Boosting (AdaBoost) technique where the output from the Machine Learning (ML) classifiers are combined to produce a single output. Adaboost combines the poor classifiers and extracts the prediction value to make a better classifier. The experimental results show that the proposed ensemble classifier provides better accuracy of 93.42 % that is comparatively better than existing Convolutional Bidirectional - Long Short-Term Memory (ConvBiLSTM) classifier and Hybrid Lexicon-Naïve Bayes Classifier (HL-NBC) which produce classification accuracy of 91.53 % and 89.61 % respectively.

1. Introduction

Twitter have about 319 million of active number of users in a month, this is considered as a treasury for the organizations and individual with strong political and economic background and keenly shows interest in preserving their reputation (Minaee et al., 2019, Naseem et al., 2020). Twitter acts as a platform to share the public opinion and their view on the organizations or individuals. Sentiment analysis acts as a key tool for the organizations to know about their capability and public views on their organization by analyzing the tweets based on the sentiment

(Gandhi et al., 2021). The analysis of sentiment in Twitter platform is emerging popular among the public where they express their feedback through texts (tweets) (Jianqiang et al., 2018). The Sentiment Analysis (SA) is considered as one of the predictable research field of Natural Language Processing (NLP) and SA plays an important role in automated detection of sentiment present in the text. Moreover, the SA is vastly availed based on reviews, surveys and economic report etc. (Akhtar et al., 2019, Soumya and Pramod, 2022). In the process of SA tweets of similar type of tweets are taken into consideration then it is categorized into words and sentences, this is known as tokenization. The language

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utilized in Twitter is universal, not in proper structure and varying slangs. So, a pre-processing step must be involved in filtering the unwanted data to enhance the Sentiment Analysis results (Das et al., 2020, Nurulhuda Zainuddin et al., 2018).

The classification models based on deep learning have emerged and attained its peak in various tasks such as answering for the questions, analysis of sentiments, classification and generation of images, analysis of sentiments (AlBadani et al., 2022). Users of Twitter post many messages throughout the day that can include thoughts about brands, organizations, things, and public relations (Khalid et al., 2020). These opinions can be divided into three different attitudes: adverse, favorable, and neutral. There are various levels that can be assigned to each attitude. This technique is known as sentiment analysis, analyses and extracts contextual knowledge from unprocessed data. (Kim and Jeong, 2019, Bibi et al., 2022, Sergiu Cosmin Nistor et al., 2021). Analysis of sentiments takes place through two stages such as extraction of features and categorization of sentiments. The extraction of relevant features helps to choose the entities in a particular topic (Sergiu Cosmin Nistor et al., 2021, Nurulhuda Zainuddin et al., 2018). Moreover, the words that change from one dataset to other results in poor prediction capacity according to the ML models, obtained using Contextual Analysis (CA). It should be emphasised that without the aid of any linguistics resources, the words are automatically clustered and significant nodes are determined. Improved prediction outcomes should be taken in to the account while performing sentimental analysis from discrete real time dataset the usage of individual classifiers does not result in better classification accuracy. So this research considered this drawback as a motivation and introduced a sentiment analysis model to analyze collected tweets from Twitter data based on positive, negative and neutral tags. Here an ensemble based classifier is proposed to classify emotion on the basis of tweets given by user. Sentiment analysis mostly looks for positive or negative expressions of opinion regarding a person, news story, or digital material. Understanding user sentiment allows businesses, service providers, and individuals to reevaluate their approaches and enhance their offerings. Additionally, it aids businesses in obtaining critical feedback regarding their goods and areas in need of development.

This research is employed by the following contributions:

1. To investigate how various pre-processing methods and subjectivity analysis affect sentiment classification performance and accuracy.
2. To extract the features that significantly influence the reviews' sentiment by using an appropriate feature extraction mechanism.
3. A feature selection approach is proposed to determine which features have the most impact. The goal of this strategy is to increase categorization effectiveness in order to get pertinent opinions.
4. A new ensemble classifier is proposed which is a combination of ML classifiers (RF, DT, SVM) and ensemble classifier performs high level classification of sentiment in tweets

Rest of the paper is organized in the following way; the literature review is provided in Section 2. The methodology of the proposed technique is shown in Section 3. The results and analysis of the is discussed in Section 4 and finally, the conclusion of the paper is presented in Section 5.

2. Related works

Sakirin Tam et.al (Tam et al., 2021) developed a combined framework of Convolution neural network and Bidirectional Long Short-Term Memory known as ConvBiLSTM which is a word embedding model that converts the tweets into numerical values. The CNN present in the framework receives embedded features as an input and provides features with small dimensions. The Bi-LSTM collects the data from the CNN layer and provides classification result. The framework utilized Word2Vec and GloVe for the process of word vectorization. The ConvBi-LSTM framework has the capability to capture both local and

global contextual sentence in Twitter. However, the word representation present in the framework affects the classification accuracy of the whole framework.

Anisha P. Rodrigues and Niranjan N. Chiplunkar (Rodrigues and Chiplunkar, 2022) developed a framework based on Hybrid Lexicon-Naïve Bayes Classifier (HL-NBC) for classifying emotion of the tweets. Here, analysis of sentiment was based on classification of topics where the tweets are classified and inappropriate tweets were filtered out. The introduced framework utilized Hadoop framework which combines Apache Flume to gather the Twitter data and perform classification. The framework based on HL-NBC effectively classifies large number of real time Twitter data and analyze the concerned tweets. However, HL-NBC framework found difficulty in classifying the tweets based on sarcastic opinions of the public.

Babacar Gaye et.al (Gaye et al., 2021) have presented a hybrid stacked ensemble technique which was utilized in classifying the tweets based on sentiments by using an ensemble based three Long Short-Term Memory (LSTM) as base classifiers and Logistic Regression (LR) as a meta classifier. In hybrid stacked ensemble technique, the tweets classification takes place with the sentiments obtained from Text Blob and original sentiments. The hybrid stacked ensemble technique was effective with less consumption of time because it doesn't need any feature extraction method. However, the suggested model does not suit for classifying of sarcastic texts, fake reviews etc.

Muhammad Umer et.al (Umer et al., 2022) have introduced an Extra Tree Convolutional Neural Network (ETCNN) based ensemble model for classification of sentiment based on tweets related to COVID-19. The ET and CNN models are combined using the methodology of soft voting. A large dataset in an unstructured form was obtained from publicly available dataset, then the data was pre-processed and annotated using TextBlob and VADER. The ETCNN based ensemble model utilized TF-IDF and Word2Vec model to perform precise classification with various ML models. However, the ETCNN based ensemble model was limited for analysis of three feature approaches and it does not fit for approaches based on annotations.

Mehmet Umut Salur and Ilhan Aydin (Salur and Aydin, 2020) have introduced a deep learning model which combines word embedding models with various deep learning models such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional Long Short-Term Memory (Bi-LSTM) or Convolutional Neural Network (CNN). The features were extracted from various deep learning models and the text was classified accordingly. The character-level embedding and FastText embedding was performed using the algorithms of CNN and LSTM. The features collected from CNN and LSTM was combined and transferred to the softmax layer for classification. Generally, deep learning model was much effective for huge datasets but in the proposed model the dataset used was minimal in size.

Jawad Khan et al (Khan et al., 2023) have introduced a Sentiment and Context Aware Attention-based Hybrid Deep Neural Network (SCA-HDNN) model along with attention mechanism which effectively figure out the salient features in the text. Initially, integrated wide coverage sentiment lexicons are used to detect the sentiment features and BERT model is used to detect the sentiment of the extracted words. Moreover, the suggested model utilized attention mechanism that allots weight to the features and finally, CNN is used to classify the sentiment of the tweets by reducing the feature dimensionalities. However, the suggested approach was not suitable to analyze the sentiment of the text at the aspect level.

Fang, X., Zhan, J. (Fang and Zhan, 2015) The authors made the study that addresses sentiment polarity categorization, a basic topic in sentiment analysis. The study uses a selection of online product reviews from Amazon.com as its data source. A proposed process for categorizing sentiment polarity has been presented, complete with step-by-step instructions. Both review-level and sentence-level categorization experiments have been conducted using machine learning classifiers.

• Rezaul Haque, Naimul Islam, Mayisha Tasneem, Amit Kumar Das

(Haque et al., 2023) For multi-class SA, a novel CNN-based LSTM network called CLSTM was proposed. It can increase the classification score by extracting more contextual data from Bengali social media texts. Additionally, it provided a performance comparison analysis between DL and ML classifiers using the CLSTM model were suggested. The team developed a web application based on the best-performing machine learning algorithm, LR, using the Flask framework. The model called CLSTM was developed to categorize real-world social media text sentiments into four groups based on their proposed model. Unveiled a multi-class dataset of 42,036 social media comments in Bengali that were divided into four groups political, sexual, acceptable, religious.

Zulfadzli Drus, Haliyana Khalid (Drus and Khalid, 2019), Their research examines relevant literature produced between 2014 and 2019 in order to better comprehend the application of sentiment analysis in social media platforms. Sentiment analysis is a method that divides opinions into positive, negative, and natural sentiments by extracting, converting, and interpreting opinions from texts using Natural Language Processing (NLP). In order to better understand their customers and make the required decisions to improve their products or services, the majority of the prior studies used sentiment analysis in product or movie reviews.

3. An ensemble based classifier to classify the sentiment of tweets

Sentiment analysis is considered as an important tool to collect the information regarding the public view in applications related to daily life. The proposed work extracts sentiment from the tweets posted by user's different situations. Moreover, ensemble based classifier has the ability to classify the emotion from text. The classification of sentiment analysis using ensemble classifier is diagrammatically represented in Fig. 1

The sentiment analysis model evaluated emotion of the tweet from Twitter API which recognize the text based on its nature. Analysis of sentiment is a significant task which helps to deliver the exact sentiment of the tweets. The proposed model has different phases such as (i) pre-processing, (ii) extraction of feature, (iii) weighted feature selection and (iv) classification.

3.1. Data acquisition and pre-processing

The raw data obtained from Twitter API dataset (Zhao et al., 2020) and SST-2 (Tam et al., 2021) dataset is pre-processed to remove large amount of unwanted data which complicates the process of classification. Twitter data is frequently inappropriate for direct analysis. So, the data should be normalized to offer an improved format before utilizing any methodologies. In the suggested methodology, pre-processing

techniques are used to eliminate unnecessary contents. As a result, the data can be used by all learning methods. One of the key steps in data preparation, sometimes referred to as data mining, is the transformation of raw data into an understandable format. There are more errors in real-world data due to its irregularity, fragmentation, and lack of exact trends or behaviors. Pre-processing the data is therefore a key strategy for resolving this issue that arises during sentimental analysis. The data pre-processing aids in improving the classification accuracy of the model. The steps involved in pre-processing is diagrammatically represented in Fig. 2 as follows

- i) **Removal of stop words:** The redundantly used words such as adverbs, prepositions, articles are neglected from the tweets. Moreover, by removing these words, the dataset's dimensionality can be reduced. The removal of stop words aids in enhancement of classification process.
- ii) **Removal of blank space:** The blank space present in between the tweets must be removed, since the unnecessary blank space consumes more time for classification.
- iii) **Removal of punctuations:** Characters such as "." and "&" is detected in the tweets of Twitter which must be removed to enhance the classification efficiency. The tags and punctuations present in the data is removed and it is provided as input for the feature selection process.

The words must be transformed into lower case since analysis of text is case sensitive. When this process is not performed, the deep learning model compute two words (for example, 'Extraordinary' and 'extraordinary') individually which may affect the performance of the classifier.

3.2. Feature extraction

The pre-processed output undergoes the stage of feature extraction where the features are separated using three types of feature extraction process such as Bag of n-gram method, word 2 vector method and Term Frequency – Inverse Document Frequency (TF-IDF) method. The brief description of these methods are provided in this section.

The bag of n-grams is used to check that continuous words exist in textual data for which the sentiment should be analyzed. This model consists of three main types: unigrams, bigrams, and trigrams. A single word is represented by a unigram, a pair of words is represented by a bigram, and three or more words are represented by a trigram. The words extracted using bag of n-gram is denoted as T_r^{SE} . The T_r^{SE} phrases that were extracted are combined with the word2vector approach to produce a word vector representation. It is possible to identify and learn the vector representation of each word. Word2vec offers the Continuous

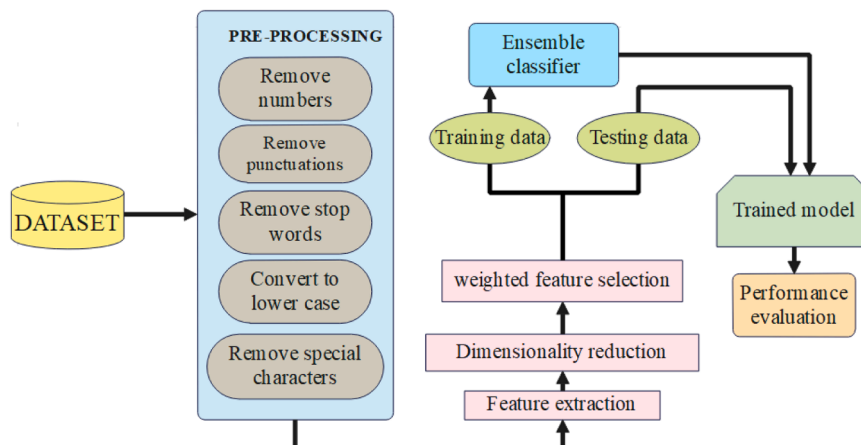


Fig. 1. Process involved in classifying the sentiment from tweets.

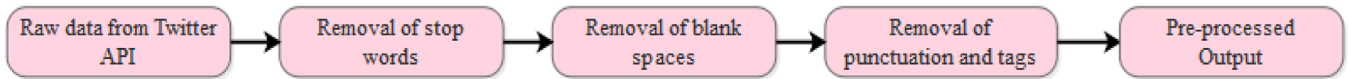


Fig. 2. Steps in pre-processing.

Bag-Of-Words (CBOW) and skip-gram models as its two training techniques. This technique determines the degree of textual similarity among the words present in the certain file. Each word identifies its originality based on a bag of words, demonstrating the lack of contextual association among two words. The extracted sentences are provided to TF-IDF, which is one of the best techniques for removing the important information from tweets. The value of TF is evaluated using the Eq. 1 as follows:

$$T = \frac{N_s}{TN} \quad (1)$$

Where the total number of terms is represented as .

The significant term is measured by utilizing IDF that specifies significant terms present in the tweets. The IDF value is represented in Eq. (2) as follows:

$$IDF = \log_e \left(\frac{ND}{TD} \right) \quad (2)$$

Where number of tweets is represented by TD and number of terms present in tweet is represented by ND . At last, the weighted term is computed using the Eq. 3.

$$TF - IDF(s, DC) = TF(s, DC) \times IDF(s) \quad (3)$$

Where s represents number of terms and DC represents text.

3.3. Dimensionality reduction using LDA

The extracted features undergo the process of dimensionality reduction where LDA focuses on creating a dataset with more characteristics and strong separability, which aids in cutting down on computing expenses. LDA supports to acquire less dimensional space while maximizing multi-class separations without compromising class information. The objective of LDA is signified using Eq. 4 – Eq. 6 represented as follows:

$$a_{opt} = \argmax \frac{a^T S_b a}{a^T S_w a} \quad (4)$$

Where,

$$S_b = \sum_{i=1}^c n_i (\mu_i - \mu) (\mu_i - \mu)^T \quad (5)$$

$$S_w = \sum_{i=1}^c \sum_{j=1}^{n_i} (x_j - \mu_i) (x_j - \mu_i)^T \quad (6)$$

Where number of classes is denoted as c , μ_i is represented as mean and μ is denoted as total mean values. The between class scatter matrix is represented as S_b and S_w is denoted as within class scatter matrix (Tam et al., 2021). By enhancing the value for Eq. 2, the LDA's efficiency will be increased. Eigenvectors a for largest Eigen values is obtained from Eq. 7.

$$S_b a = \lambda S_w a \quad (7)$$

In categorization of text, the issues arise in matrix S_w , since group of terms in text is higher when related to total texts.

3.4. Feature selection

The features obtained after the process of dimensionality reduction using LDA undergoes the stage of feature selection. The process of reducing the amount of input variables while creating a model is known as feature selection. By limiting the amount of input variables, feature selection improves the model overall performance while also lowers the computing costs. By neglecting the redundant information, the number of input variables are lowered. The three types of feature selection algorithms are filter technique, wrapper method, and embedding method. Before applying a classification algorithm, the filter method performs automatically to ignore features that look out of place. In the paper, the characteristics with high calculated values are taken into account and the features with less values are ignored.

The Wrapper technique is utilized for the development of feature selection in selecting the appropriate features. The iterative wrapper approach chooses a subset of characteristics with each iteration. The number of iterations rises whenever the dataset has a high rate of dimensional data. Wrapper approach select outstanding features and provide excellent accuracy rates. Each feature in embedded type feature selection model has an evaluation score. The high features are considered, and the features with low score is neglected. The wrapper method requires less cost for computation and chooses exceptional features to provide enhanced accuracy rate when related with existing techniques.

3.5. Classification using ensemble method

The selected features using Wrapper based technique is proceeded for classification using ensemble classifier. Ensemble based method is a collection of algorithms which is utilized for classification purpose. In the paper, existing ML classifiers are combined and classification is performed using adaptive boosting method. An ensemble method was utilized for classifying the tweets based on their sentiments. Here, Ensemble method is a combination of Random Forest (RF), Decision Tree (DT), and Support Vector Machine (SVM) classifiers. Generally, ensemble based method exhibits better accuracy value when compared with the existing algorithms. The process involved in Ensembling model is diagrammatically represented in Fig. 3 as follows

3.5.1. Random Forest (RF)

Random Forests (RF) is a well-known method which is utilized for the classification and regression problems. It has developed into a method to ensemble learning that is built on several decision trees. A large number of decision trees are built during training period by RF

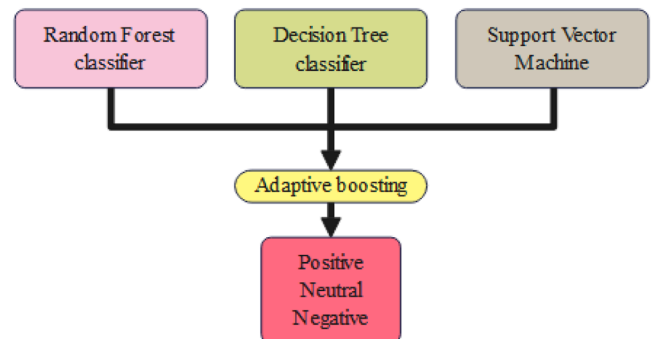


Fig. 3. Process involved in Ensemble model.

classifiers, which then provide a representation mode for classification. By decreasing the correlation between randomly chosen trees, this classifier can address the issue of overfitting and aids in boosting prediction accuracy. In RF classifier with high dimensional feature space provides less accuracy, so more number of random trees must be generated. The general procedure to construct the RF model with training data D is mentioned in Eq. (8).

$$D = \left\{ \left(f_i, C_i \right)_{i=1}^N \mid f_i \in \mathbb{R}^F, C_i \in \{1, 2, \dots, c\} \right\} \quad (8)$$

Where the features are represented as f_i , the set of classes and the count of samples used in training is represented as C_i and N respectively. Each decision tree present in the RF classifier contains set of bagged samples $D_1, D_2, \dots, D_p$. The prediction of RF approach is defined as follows:

$$\hat{C} = \text{majority} \left\{ \hat{C}_i^p \right\}_1^p \quad (9)$$

Where the prediction class is represented as $\hat{C}$.

In this paper, RF is a significant classifier which provides an outstanding accuracy in the process of classifying the sentiments. The RF classifier consist of hyper parameters such as number of trees, features, tree depth acts as a significant parameter which helps in maintain better accuracy.

3.5.2. Decision Tree (DT)

The DT classifier performs based on categorizing the region into multiple sub parts. DT classifier make use of every possible solution to provide significant interaction among the variables. The classes present in DT consist of two different categories of similar classes such as purity class and impurity classes. The impurity class present in DT must be rectified using entropy concept. The performance of DT is based on degree of the particular degree and this can be evaluated using Eq. (10) as follows:

$$H = - \sum p(x) \log p(x) \quad (10)$$

Where the probability of the element id represented as $p(x)$.

3.5.3. Support Vector Machine (SVM)

SVM accomplishes binary classification for linear and non-linear versions. In general, the datasets are in linearly closed form, so main objective of SVM is to capture the available surfaces and categorize the samples into positive, neutral and negative on the basis of minimization principal. SVM classifiers define the boundaries in a high dimensional feature space. The hyper-plane classifies the vectorized text into three categories such as positive, neutral and negative. The issues related to optimization can be reduced using the Eq. (11) as follows:

$$\vec{\alpha}^* = \text{argmin} \left\{ - \sum_{i=1}^n \alpha_i + \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \left\langle \vec{x}_i, \vec{x}_j \right\rangle \right\} \quad (11)$$

Where the feature vector is represented as x and the dimension of the vector is denoted as d . The value of y lies among the range from $\{1, -1\}$.

The SVM classifier has the capability to categorize the dataset in an individual hyper plane and in non-linear dataset kernel functions are utilized in a high dimensional spaces.

3.6. Ensemble model

Ensemble methods are familiar in machine learning and recognition of patterns and the ensemble methods combine the output of the weaker algorithms to enhance the accuracy and classification efficiency of the model. Ensemble methods are generally categorized into two types such as dependent frame and independent frame. In dependent framework, the output of the individual inducer affects the next process and the

independent inducer is constructed independently which is differed from other inducers. The ensemble model exhibits better classification accuracy because it combines the classification algorithms which is utilized to perform effectively. This method evaluates the prediction efficiency of the individual classifiers and combines it using adaptive boosting technique. The AdaBoosting technique is less prone to overfitting, since the parameters are not jointly optimized. So, this research utilized AdaBoost technique for the process of Ensemble which aids in better classification accuracy.

3.6.1. Adaptive boosting (AdaBoost) algorithm

The adaptive boosting is a boosting technique which is created for the purpose of binary classification. AdaBoost algorithm uses the concept of boosting which is ensemble to generate a robust classifier from the weaker classifiers. AdaBoost has capability to improve overall efficiency of ML classifiers. Adaboost combines the poor classifiers and extracts the prediction value to make a better classifier known as ensemble classifier. The process of combining multiple classifiers along with the training set produce weight in the final stage of voting. Ada-boost technique considers two techniques such as training the subset with weak classifier and a random subset must be utilized for training the whole group. Secondly, a weight factor must be assigned to the subset. The classifier with 50 % accuracy provides zero weight and classifier with accuracy less than 50 % exhibits negative weight for the classifier.

$$H(X) = \text{SIGN} \left(\sum_{t=1}^t \alpha_t h_t(x) \right) \quad (12)$$

Where the output of the classifier for input x is represented as $h_t(x)$ and the weight assigned to the classifier is denoted as α_t .

The value of α_t can be computed using the formula mentioned in Eq. (13) as follows

$$\alpha_t = 0.5 \times \ln \left(\frac{1-E}{E} \right) \quad (13)$$

Where the error rate produced is denoted as E .

4. Results and analysis

The overall efficiency of the developed ensemble method is evaluated and the comparison is done with existing methodologies described in related works.

4.1. Datasets

Twitter API dataset consist of tweets regarding users, spaces, direct messages, lists, trends, media, places. Twitter API is one of the public accessible dataset that contributes with the applications regarding real world. Moreover, Twitter API dataset consist of tweets with long sentences, stop words, misspelled words etc. To remove these a pre-processing technique must be performed to ease the process of classification. Secondly, the data is collected from binary labeled Stanford Sentiment Treebank (SST-2) which is comprised with data based on training, testing and validation with 67,349, 872 and 1821 sentences respectively.

4.2. Experimental setup

The system utilized in the classification process has i5 processor at 3.40 GHz on Ubuntu 16.4 OS with 8 GB RAM. The experiment is processed for the tweets from Twitter API dataset. The performance of the classifier is evaluated based on accuracy, precision, recall and f-1 score.

4.3. Performance metrics

The performance of the ensemble classifier in classification of sentiment is evaluated based on accuracy, precision, recall and F1 score.

Accuracy: It is stated as the total count of properly categorized data to total count of data.

$$\text{Accuracy} = \frac{TN + TP}{TN + FP + TP + FN} \quad (14)$$

Where, TN denotes true negative, TP denotes true positive, FP denotes false positive and FN denotes false negative.

Precision: It is the total number of truly positive values to the total number of positives.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (15)$$

Recall: It is defined as the ratio among the total count of Positives that is properly categorized as Positive to total samples of positive and negative values.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (16)$$

F1-score: It is evaluated by using both recall and precision. when values of precision and recall are 1, the F1 score will be one.

$$\text{F1 score} = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

4.4. Performance analysis

The proposed ensemble classifier is a combination of SVM, DT and RF. The performance of the proposed ensemble classifier is compared with the individual performance of SVM, DT and RF. Table 1 represented below represents the performance of the proposed ensemble classifier in classifying the sentiment of tweets.

Results from Table 1 and Fig. 4 shows that the ensemble classifier attained better accuracy of 93.42 % during classification that is comparatively higher than existing ML classifiers. The better result is due to the capability of ensemble classifier in combining the individual predictions of RS, SVM and DT. An ensemble classifier combines the predicted values and provide it for the adaptive boosting technique which uses the concept of boosting which is ensemble to generate a robust classifier from the weaker classifiers. AdaBoost has capability to improve the overall efficiency of ML classifiers.

4.5. Statistic analysis

It is evaluated for ensemble classifier to evaluate the performance of proposed model with diverse classification models. The statistical analysis is processed by considering the statistical metrics such as best, worst, mean and standard deviation. The Table 2 and Fig. 5 presents the statistical graph representation of the proposed classifier with existing classification approaches.

The results from Table 2 and Fig. 5 shows that the proposed ensemble classifier effectively classifies the statistical value of tweets by evaluating mean, median, worst and best values. By using ensemble classifier, the classification process has been done effectively with less complexities Fig. 6.

Table 1

Performance of different classifiers.

Classifiers	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
RF classifier	91	87	80	81
DT classifier	89	83	85	84
SVM classifier	86	89	78	83
Ensemble classifier	93.42	91.29	89.65	87.32

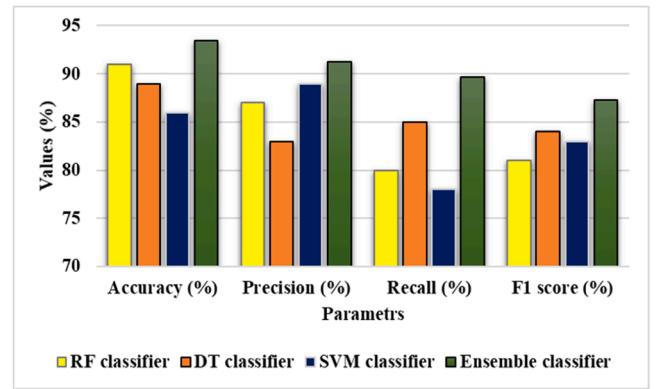


Fig. 4. Graphical representations for performance of classifiers.

Table 2

Statistical analysis.

Classifiers	RF classifier	DT classifier	SVM classifier	Ensemble classifier
Best	0.932	0.945	0.952	0.971
Worst	0.921	0.926	0.927	0.963
Mean	0.942	0.939	0.938	0.965
Median	0.936	0.934	0.936	0.947
Standard deviation	0.008	0.009	0.007	0.005

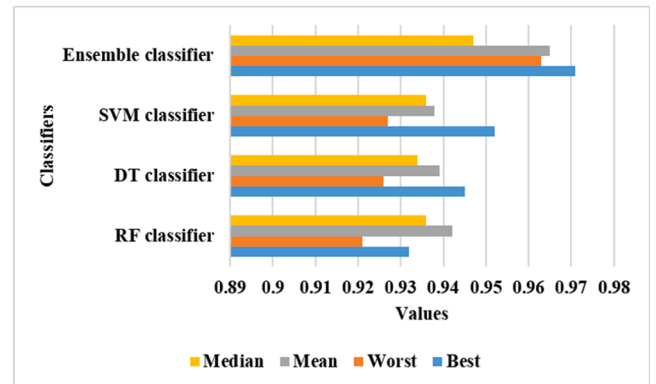


Fig. 5. Graphical representation for statistical analysis.

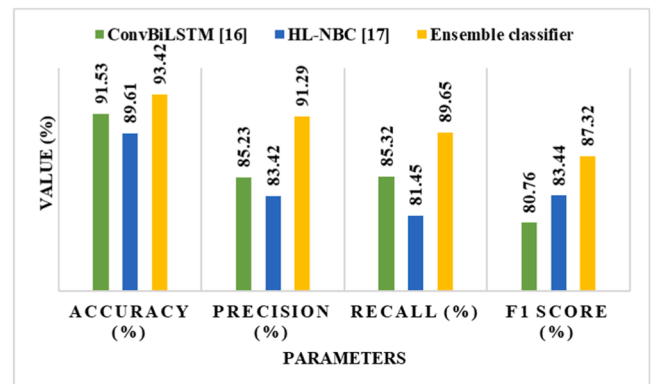


Fig. 6. Graphical representations for comparison of classifiers for Twitter API.

4.6. Performance evaluation for SST-2 dataset

In this section, the performance of the ensemble classifier is evaluated with existing classifiers such as RF, DT and SVM classifiers. The data is collected from the SST-2 dataset is used to evaluate the performance of the proposed classifier with existing classification methods. The results obtained from the evaluation of classifier with the existing approaches is presented in Table 3 as follows:

The results from the Table 3 shows that the proposed ensemble classifier have achieved better classification results. The accuracy of sentiment classification for ensemble classifier is 96.53 % which is comparatively higher than the existing classifiers such as RF, DT and SVM with 94.64 %, 91.56 % and 93.71 % respectively. The Fig. 7 shows the graphical representation of evaluation of classification for SST-2 dataset Table 4.

4.7. Comparative analysis

This section provides the comparative analysis of the proposed ensemble classifier with the existing Convolutional Bidirectional - Long Short-Term Memory (ConvBiLSTM) classifier (Tam et al., 2021), HL-NBC (Rodrigues and Chiplunkar, 2022) and SCA-HDNN with BERT (Khan et al., 2023). Table 4 represents the comparison of the proposed ensemble method with the existing classifiers described in related works.

The results from Table 4 shows that ensemble classifier attained better performance in overall metrics. The results are evaluated from Twitter API and SST-2 datasets, the proposed ensemble classifier attained better accuracy of 96.53 % for Twitter API dataset that is comparatively higher than ConvBiLSTM (88.47 %) and HL-NBC (89.61 %). Similarly, for SST 2 dataset, the proposed approach achieves classification accuracy of 96.53 % which is comparatively higher than the existing ConvBiLSTM with the accuracy percentage of 88.47 and SCA-HDNN with accuracy of 92 %. The better result of the proposed approach is due to the process of Ensembling using adaptive boosting technique where the prediction of best values from the individual features of SVM, RF and DT is provided in adaptive boosting technique to get better classification output.

5. Conclusion

Dependence on social media is unavoidable because it helps the users to know about the day-to-day activities takes place in the world. Specifically, Twitter relies as a widely used platform which helps the people to share their opinion on different topics. The sentiment analysis of tweets is a challenging process because Twitter consist of vast number of data. So, this research introduced an ensemble classifier with BERT mechanism to classify the sentiment of the tweets based on their polarities. This research used Wrapper based technique for selecting the relevant features and the Wrapper technique evaluates score for each feature, the feature with low score is ignored and high score is proceeded for classification using ensemble classifier. The proposed ensemble classifier which is a combination of ML classifiers such as SVM, RF and DT. The individual prediction values are gathered from the ML classifiers are fed to the AdaBoost technique that combines the poor classifiers and extracts the better prediction value to make a better classifier. AdaBoost has capability to improve the overall efficiency of ML classifiers. The proposed ensemble classifier effectively classifies the sentiment on the basis of paragraph-level for long tweets. The developed ensemble model achieves better accuracy of 93.42 % which is comparatively better than existing ConvBiLSTM classifier (91.53 %) and HL-NBC classifier (89.61 %). However, when the number of levels get increased in DT, the model become vulnerable and leads to high variance which effects overall efficiency of the ensemble classifier. In future, the proposed ensemble classifier can be utilized to perform sentiment analysis in different social platforms.

Table 3

Evaluating the classification performance for SST-2 dataset.

Classifiers	Accuracy (%)	Precision (%)	Recall (%)	F-1 score (%)
RF classifier	94.64	89.93	90.06	90.21
DT classifier	91.56	92.82	88.13	89.99
SVM classifier	93.71	91.39	85.32	88.62
Ensemble classifier	96.53	95.87	91.24	93.91

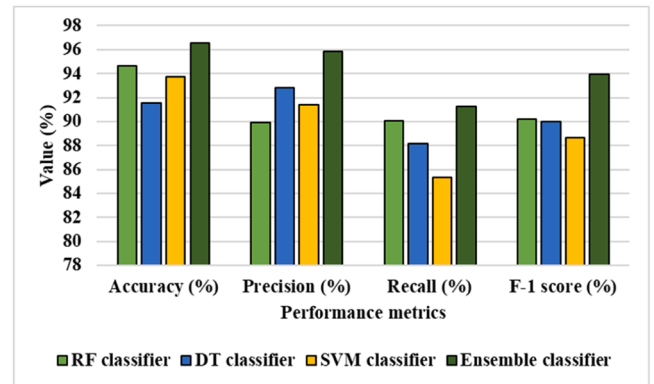


Fig. 7. Graphical representation for evaluating the classification performance for SST-2 dataset.

Table 4

Comparative results for Twitter API and SST-2 dataset.

Classifiers	Datasets	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
ConvBiLSTM (Tam et al., 2021)	Twitter API	88.47	83.39	84.78	77.90
	SST-2	91.53	85.23	85.32	80.76
HL-NBC (Rodrigues and Chiplunkar, 2022)	Twitter API	89.61	83.42	81.45	83.44
	SST-2	92.0	90.8	93.0	91.9
SCA-HDNN With BERT (Khan et al., 2023)	SST-2	92.0	90.8	93.0	91.9
Ensemble classifier	Twitter API	93.42	91.29	89.65	87.32
	SST-2	96.53	95.87	91.24	93.91

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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